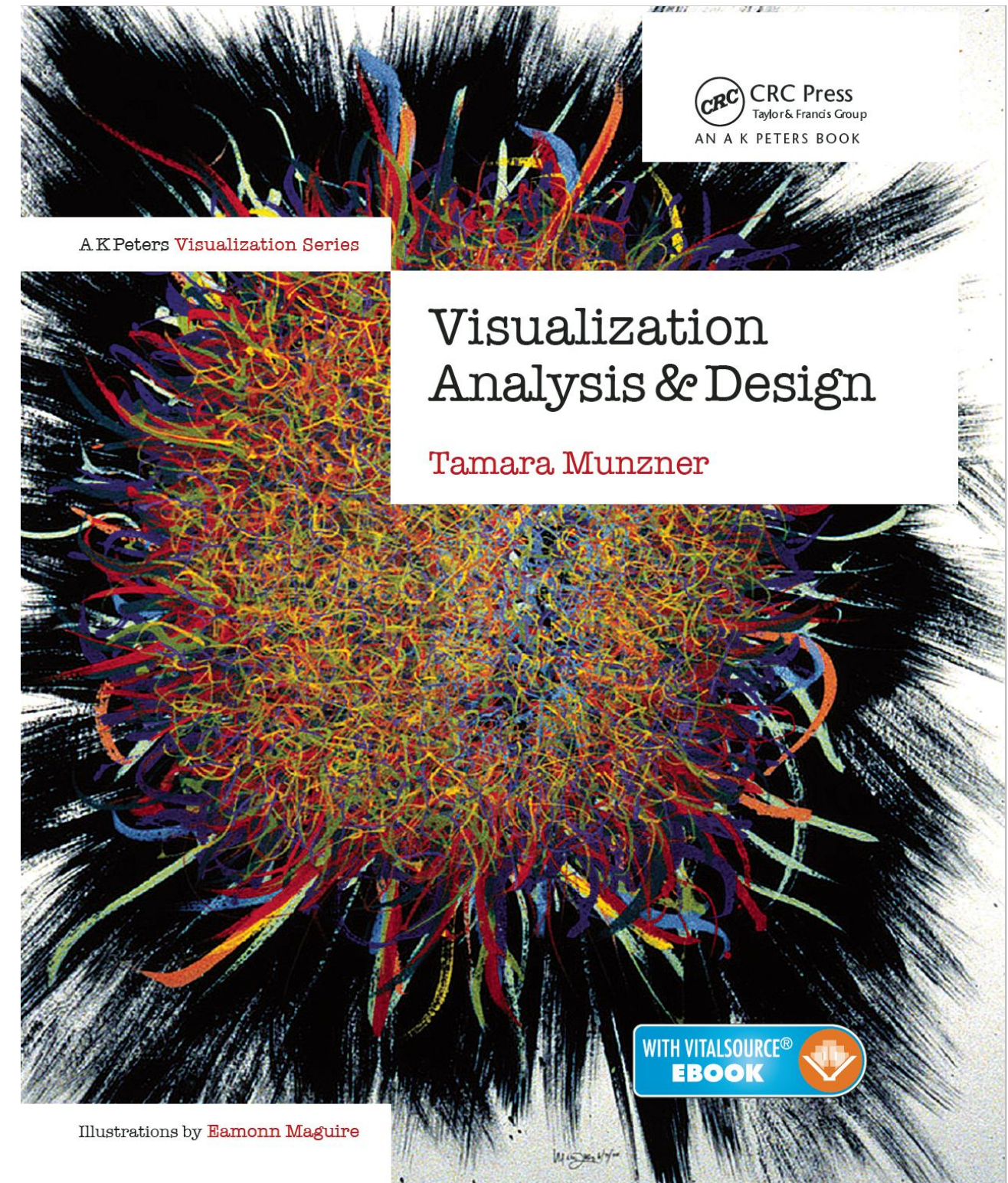


CDS505

Data Visualization & Visual Analytics

Manipulate View

*Nur Intan Raihana Ruhaiyem, PhD.
School of Computer Sciences
Universiti Sains Malaysia*

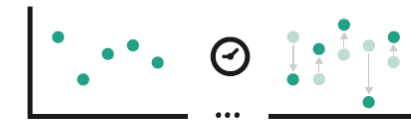


Why we manipulate view?
Or, what is the **purpose** of view manipulation?

- ✓ Introduction
- ✓ Change over time
- ✓ Select and highlight
- ✓ Navigate: Changing Viewpoint
- ✓ Navigate: Reducing Attributes

- **Change over time** - change idioms, and any of the parameters for a particular idiom. *Any aspect of visual encoding (ordering, spatial arrangement, colour), filtering of elements, detail of aggregation, and partitioning into multiple views.*
- **Select** - A change often requires a set of items or attributes: *what kind of elements, how many elements and how to highlight the elements.*
- **Navigation** - change of viewpoint; i.e., changing what is shown in a view based on the metaphor of a camera looking at the scene from a moveable point of view.

③ Change over Time



③ Select



③ Navigate

→ Item Reduction

→ Zoom

Geometric or Semantic



→ Pan/Translate

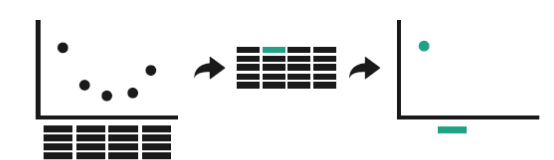


→ Constrained



→ Attribute Reduction

→ Slice



→ Cut



→ Project

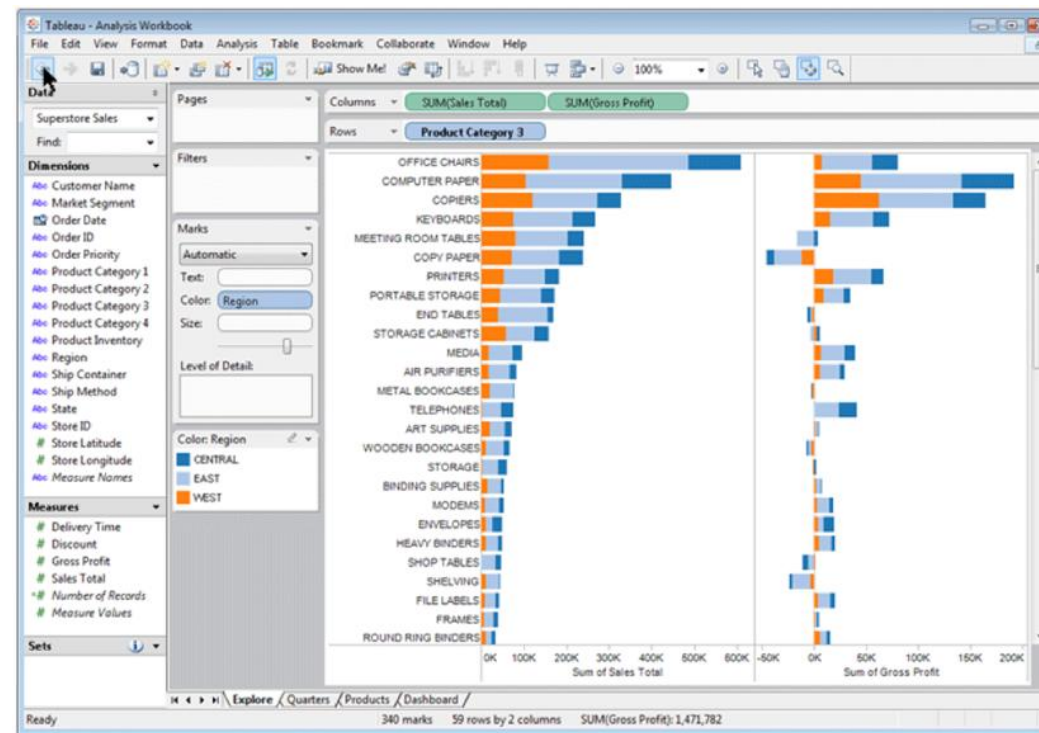
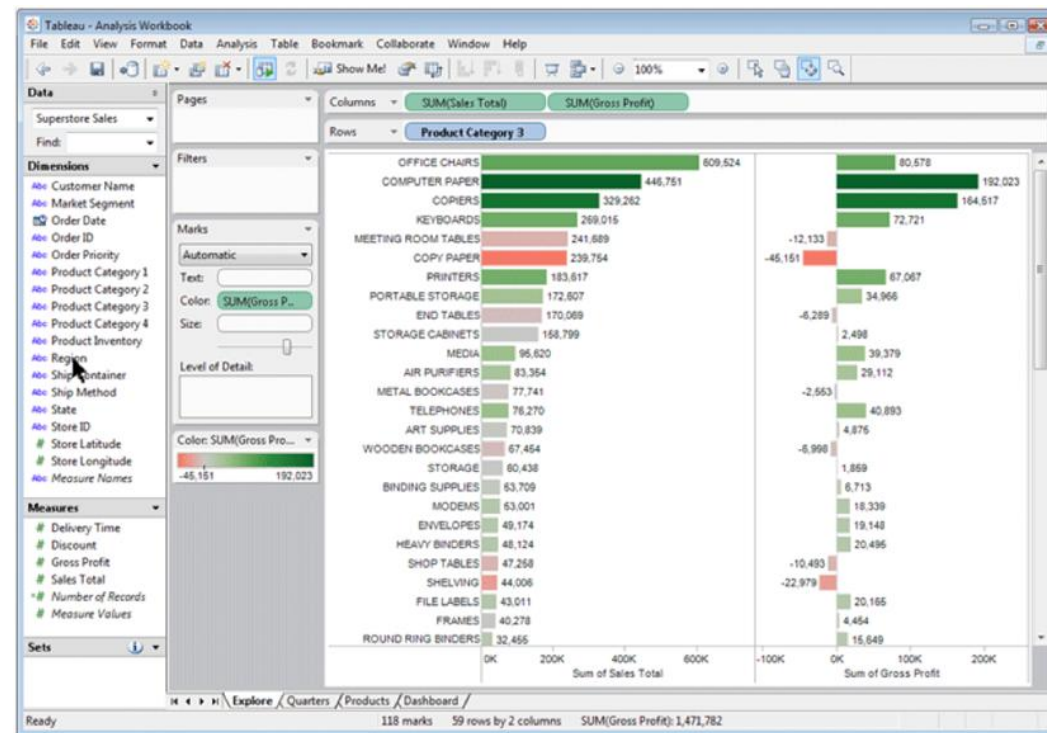


Figure 11.1: Design choices for idioms that change a view

Change any of the other choices

- encoding itself
- parameters
- arrange: rearrange, reorder
- aggregation level, what is filtered
- interaction entails change

Change View Over Time



Idiom: Re-encode

Tableau supports fluid changes between visual encoding idioms with drag and drop interaction.

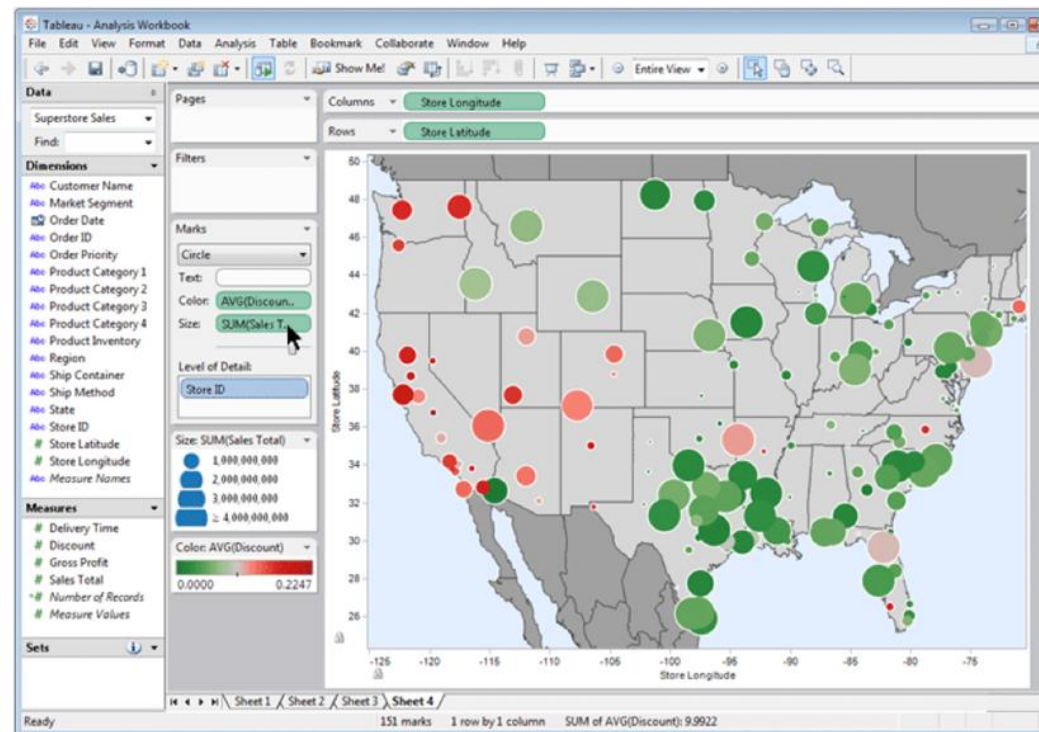
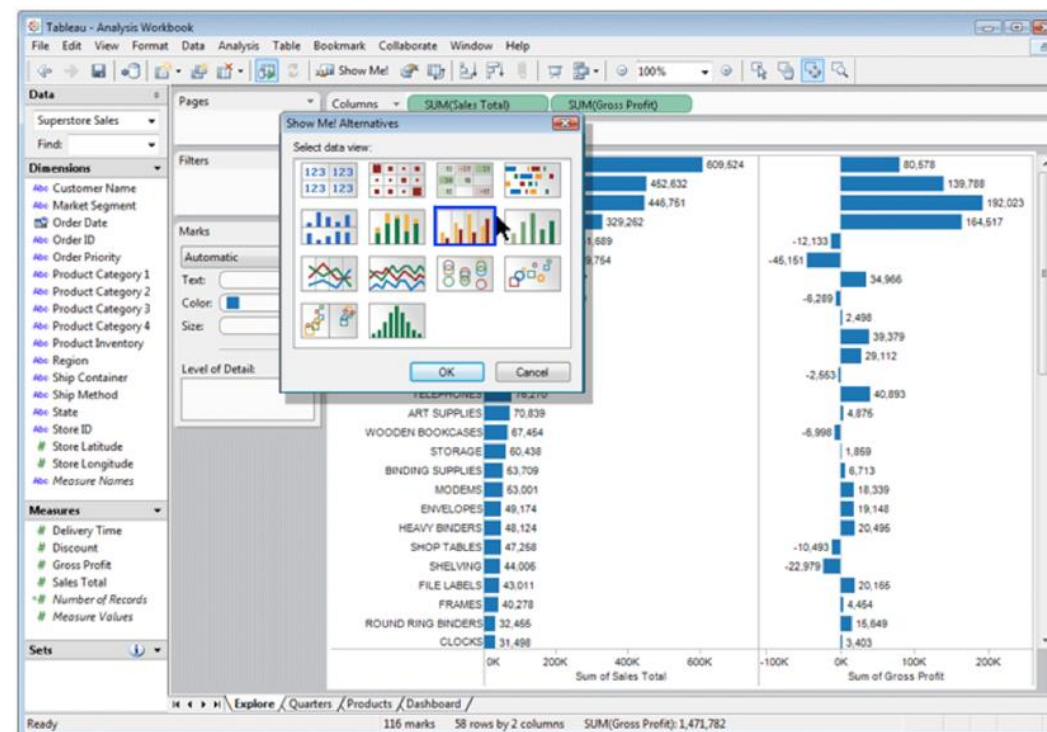


Figure 11.2: Tableau supports fluid changes between visual encoding idioms with drag and drop interaction. (a) Data encoded with bars. (b) Data encoded with stacked bars. (c) The user selects a completely different visual encoding. (d) Data encoded using geographic positions.

Change View Over Time

Idiom: Re-order

- Data: tables with many attributes
- Task: compare rankings

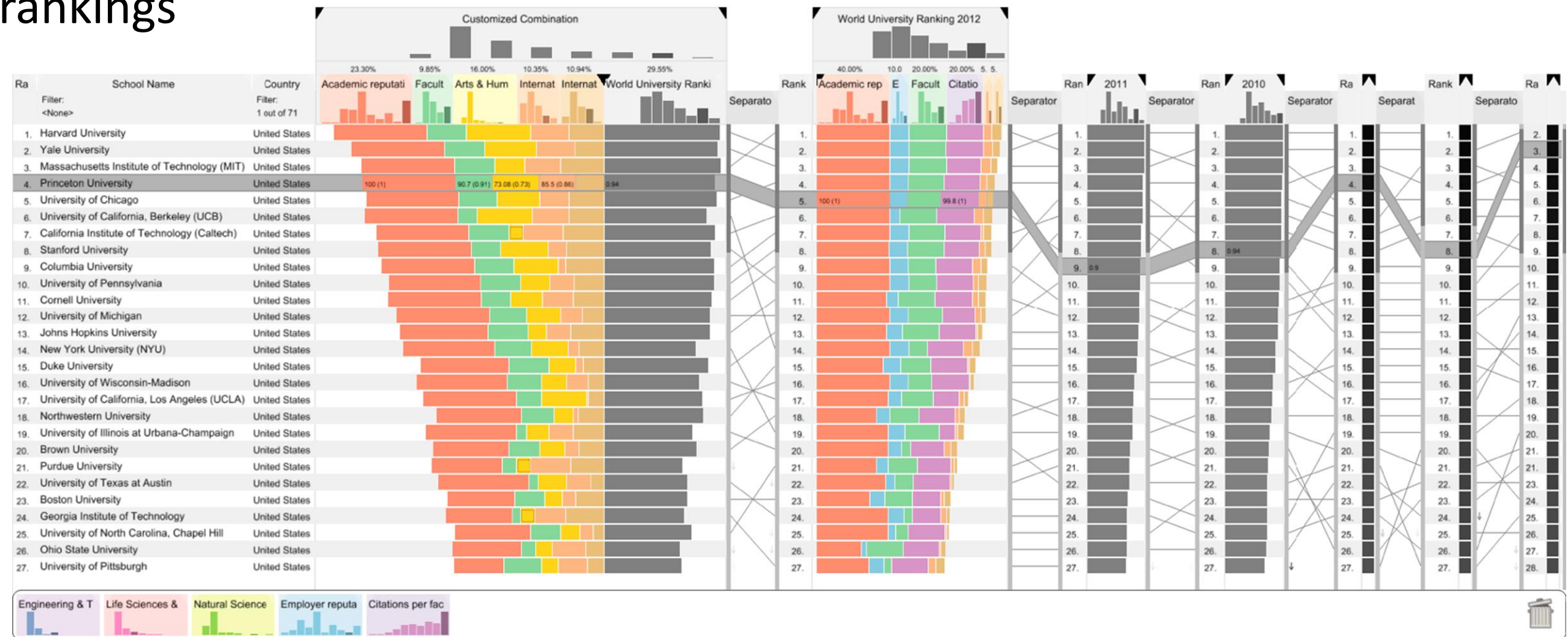
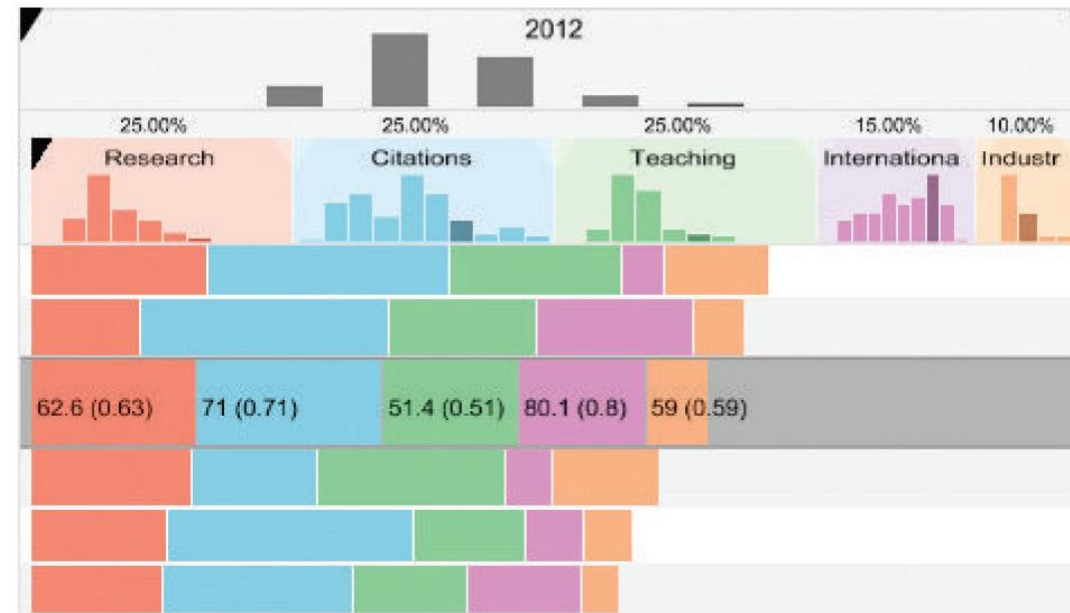


Figure 11.3: From [Gratzl et al. 13] rankings of top universities. (a) Left - a customized combination of attributes and weights for the 2012 data. (b) middle - official ranking for 2012 (coloured stacked bar charts). (c) The next two columns - a compressed view for years 2011 and 2010, (d) the last three columns - a collapsed heatmap view (grayscale)

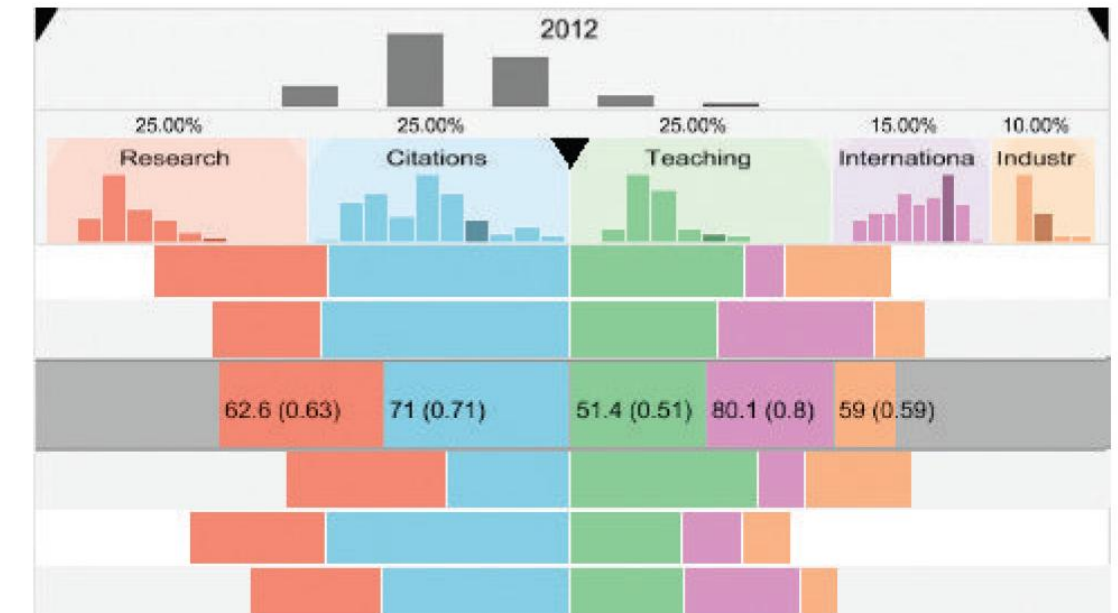
Change View Over Time

Idiom: Realign

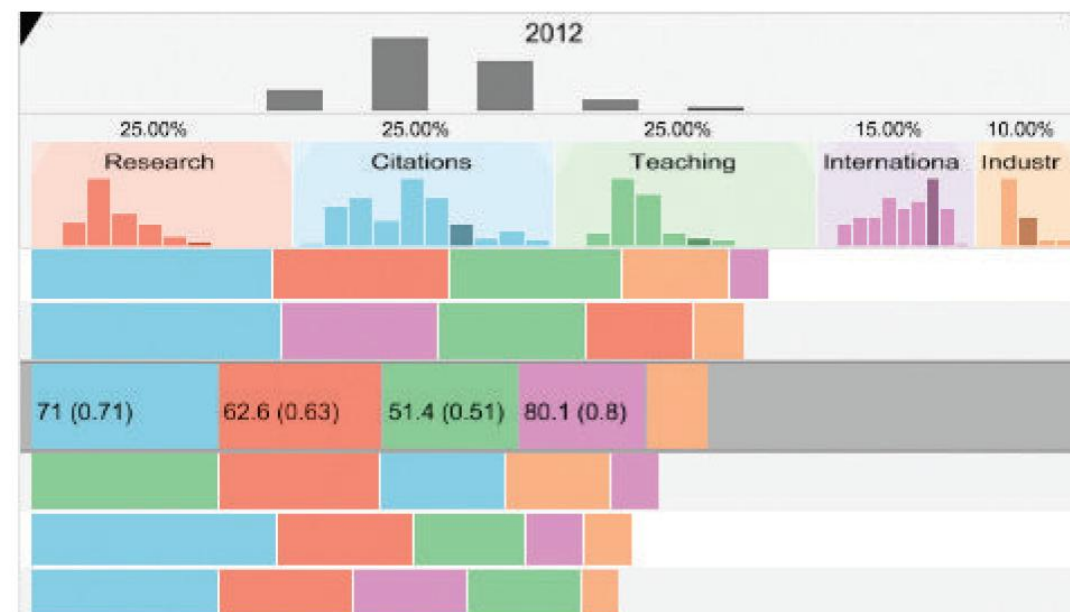
- stacked bars - easy to compare
- align to different segment - supports flexible comparison



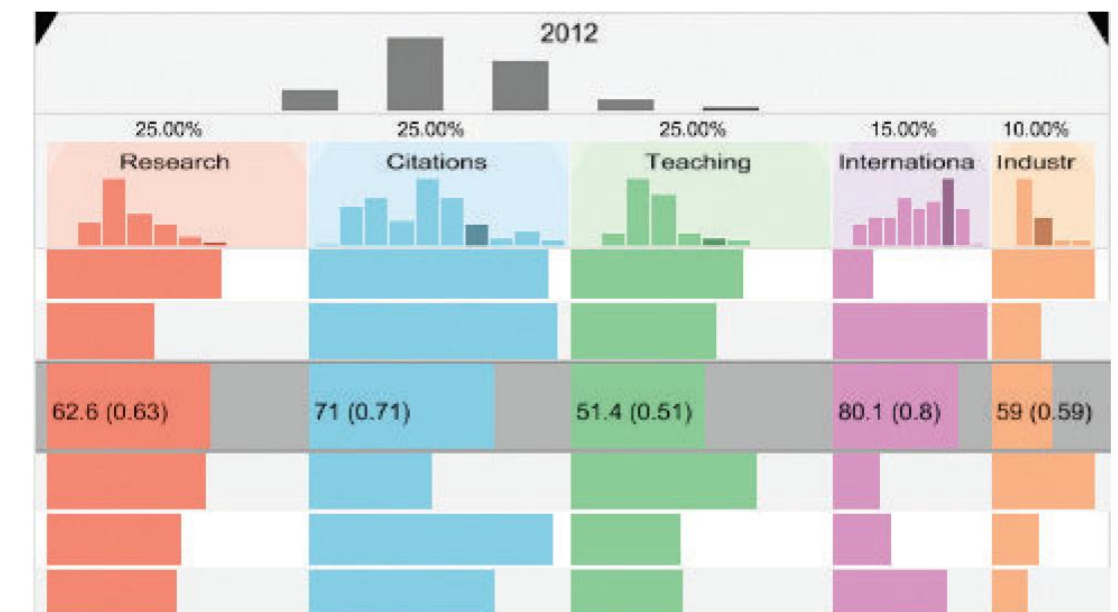
(a)



(b)



(c)



(d)

Figure 11.4: Changing the alignment in Lineup. (a) Classical stacked bars. (b) Diverging stacked bars. (c) Ordered stacked bars. (d) Separately aligned bars: small multiple bar charts. From [Gratzl et al. 13]

Idiom: Animated Transitions

- smooth transition from one state to another
 - alternative to jump cuts (abrupt change)
 - support for **item tracking** when amount of change is limited
- example: **multilevel matrix views**

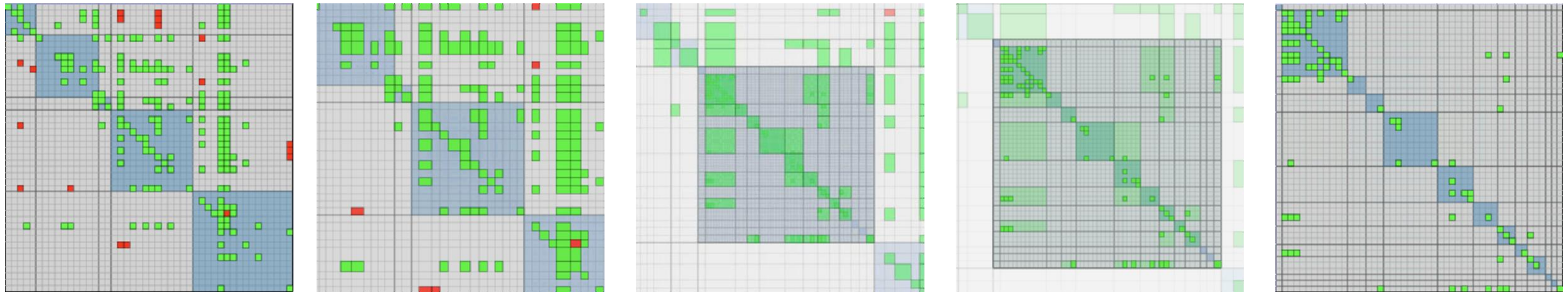
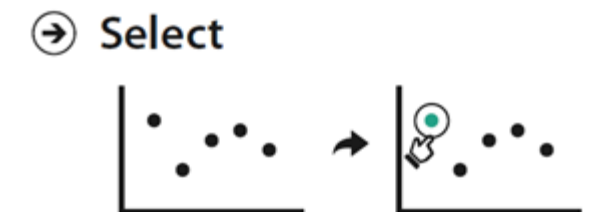


Figure 11.5: Frames from an animated transition showing zoom between levels in a compound network arranged as an adjacency matrix. From [van Ham 03]. Scope of what is shown narrows down to show only a subset of what is visible in the first frame. The middle block gradually stretches out to fill all of the available space

Select and Highlight

- **Selection** (one or more elements): basic operation for most interaction.
- **Design choices**: what kinds of elements can be selection targets. The most common element is data items, links in the case of network data. data attributes are also selectable (commonly select by levels).
- When the data is faceted across multiple views, then a view itself can also be a selection target.
- How many **selection types**?
 - click vs hover: heavyweight, lightweight
- How many **elements** can be in the selection set? **Primary vs secondary**.
- Highlight: change visual encoding for selection targets.
- **Colour - limitation**: existing colour coding hidden.
- Mark with an outline.
- Other channels e.g. motion, add explicit connection marks between items.



Select and Highlight

Idiom: Context-Preserving Visual Links

- Links are drawn as curves that are carefully routed between existing elements

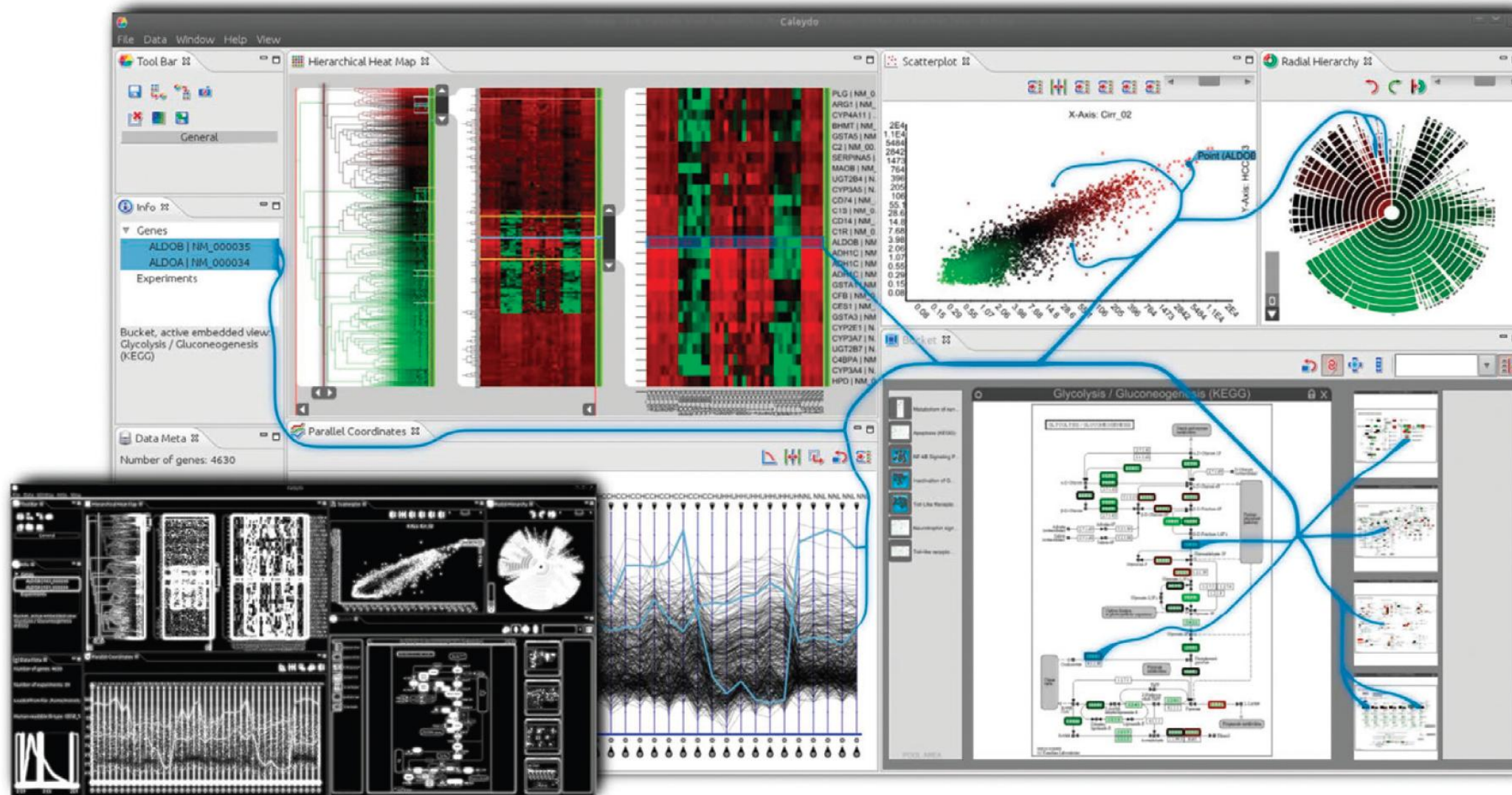


Figure 11.6: An example of the design choice to coordinate between views by explicitly drawing links as connection marks between items and regions of interest. From [Steinberger et al. 11]

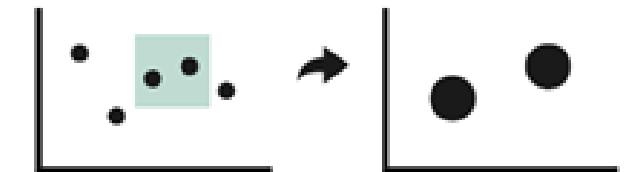
- change viewpoint
 - changes which items are visible within view
 - camera metaphor
 - Zoom
 - geometric zoom: familiar semantics
 - semantic zoom: adapt object representation based on available pixels
 - dramatic change, or more subtle one
 - Pan/translate
 - Rotate - especially in 3D
 - constrained navigation
 - often with animated transitions
 - often based on selection set

➞ Navigate

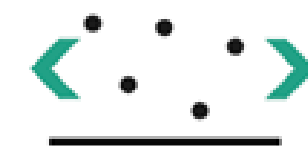
➞ Item Reduction

➞ Zoom

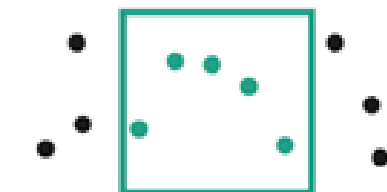
Geometric or Semantic



➞ Pan/Translate



➞ Constrained

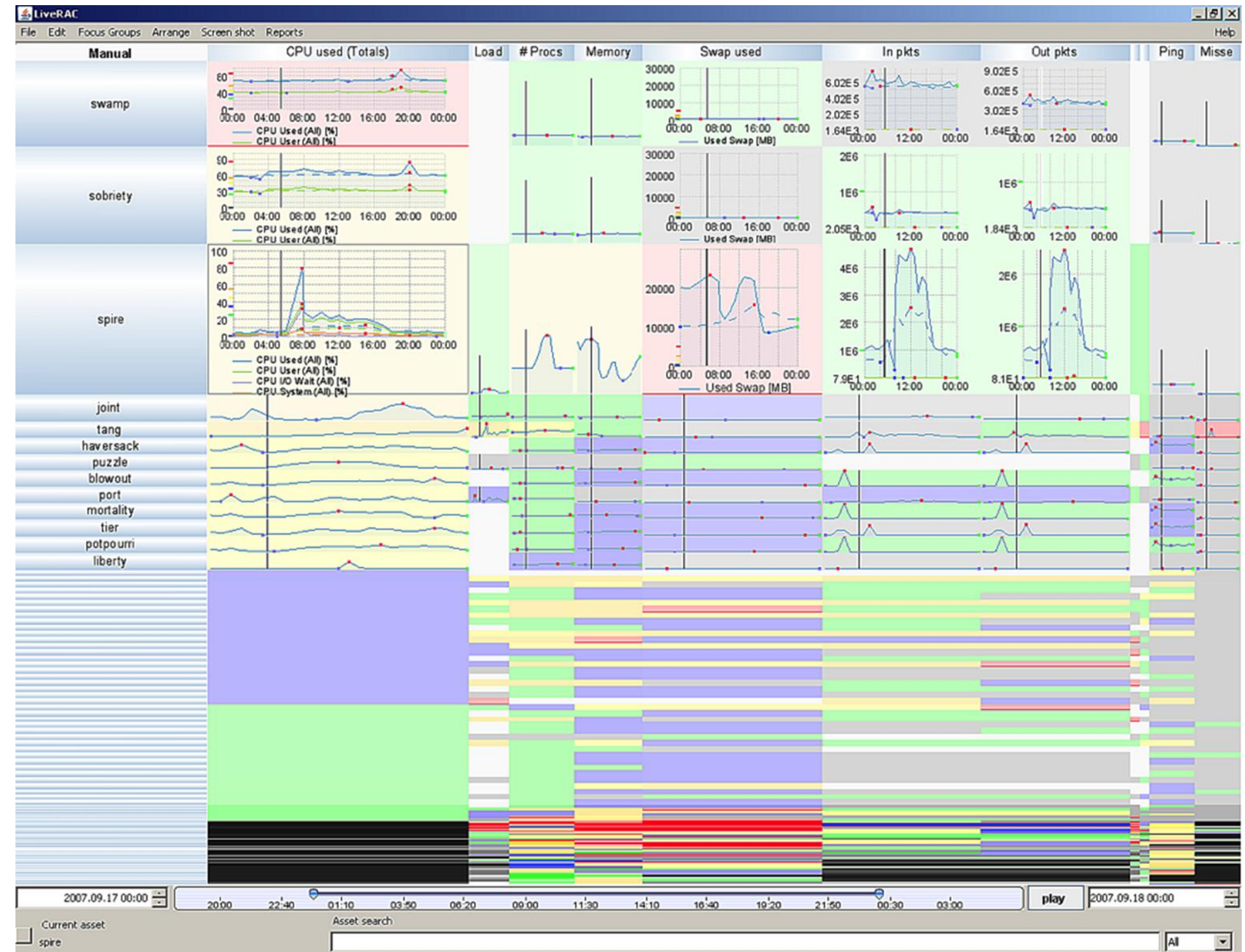


Navigate: Changing Viewpoint

Idiom: Semantic zooming

- Visual encoding change
 - coloured box (tiny box)
 - Sparkline (slightly larger)-
very concise line charts
 - simple line chart
 - full chart: axes and tick marks

Figure 11.7: LiveRAC uses semantic zooming within a stretchable grid of time series line charts. From [McLachlan et al. 08]



Continuation of camera metaphor.

- **slice**
 - show only items matching specific value for given attribute: slicing plane
 - axis aligned, or arbitrary alignment
- **cut**
 - show only items on far side of plane from camera
- **project**
 - change mathematics of image creation
 - orthographic
 - perspective

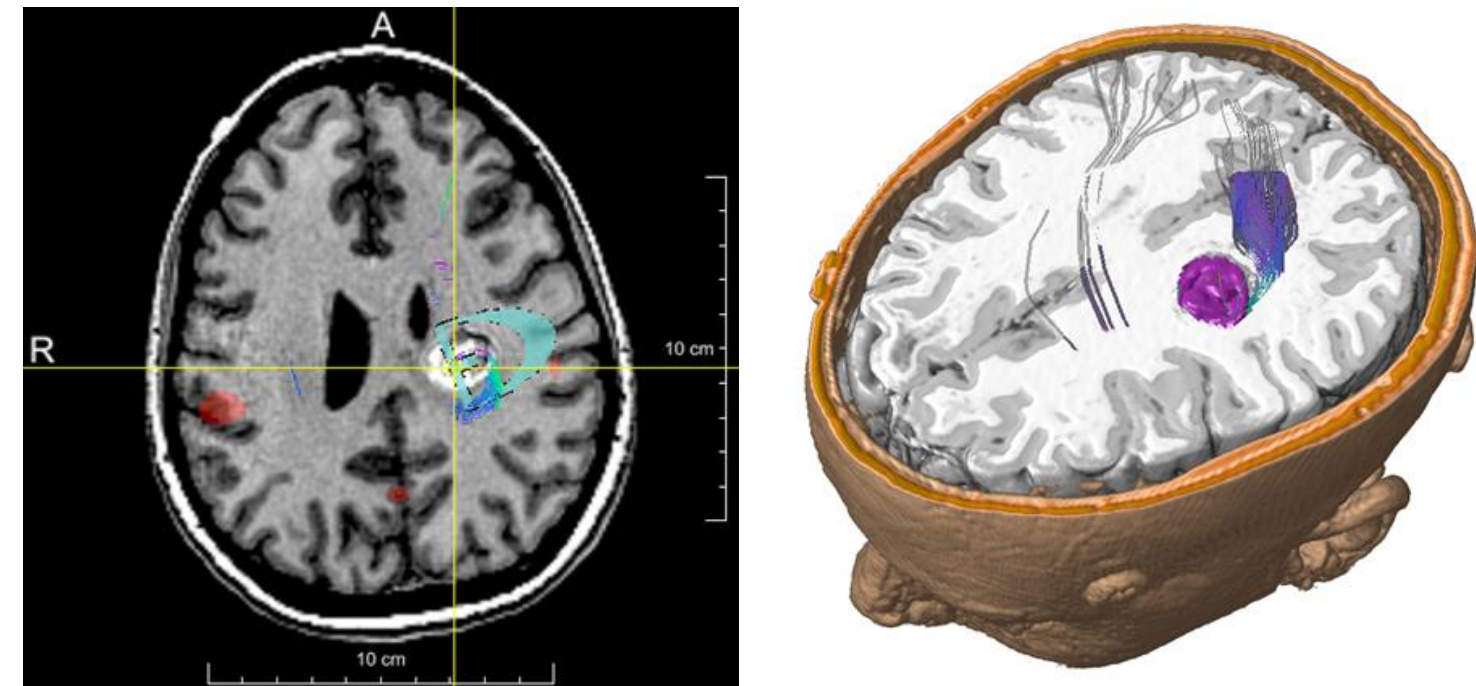


Figure 11.8: The *slice* choice eliminates dimension/ attribute by extracting only the items with a chosen value in that dimension. The *cut* choice eliminates all data on one side of a spatial cutting plane. (a) Axis-aligned slice. (b) Axis-aligned cut. From [Rieder et al. 08, Figures 7c and 0].